

Jared Reiling | Curriculum Vitae

Department of Computational Mathematics, Science and Engineering

Michigan State University

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Education

Michigan State University

Ph.D. Computational Mathematics, Science, and Engineering

Michigan, U.S.

2023-Present

Augustana College

B.A. Mathematics & Piano Performance, Summa Cum Laude

Illinois, U.S.

2019-2023

Research

Michigan State University

Graduate Research Assistant

Michigan, U.S.

September 2023-Present

Advisor: Mengsen Zhang, Ph.D.

Description: Developing a computational-experimental framework to create multiscale predictive models of naturalistic behavior and brain dynamics by integrating concepts and techniques from nonlinear dynamics, topological data analysis (TDA), and machine learning. This framework will be applied to video recordings of naturalistic social interaction in humans and animals (ferrets), and simultaneously recorded brain activity (electrophysiology). Thus, new computational frameworks are needed to model such complex neural and behavioral dynamics and to connect them across scales.

Carolina Center for Neurostimulation

National Science Foundation Research Assistant

North Carolina, U.S.

May 2024-July 2024

Advisor: Flavio Frohlich, Ph.D.

Description: Assembled and validated brain activity recording (electroencephalography) and physiology recording (electrocardiogram and respiration belt) technology for clinical trial to measure brain synchronization between a therapist and patients experiencing depression and anxiety symptoms. Additionally developed algorithms for animal (ferret) motion tracking using deep neural networks to study sustained attention during a directed task.

Baylor College of Medicine

Undergraduate Research Assistant

Texas, U.S.

May 2022-August 2022

Advisor: François St-Pierre, Ph.D.

Description: Implemented machine learning models to optimize protein directed evolution for genetically encoded voltage indicators that measure brain activity. Developed a pipeline that transfers laboratory's past and future experimental data into a database designed for machine learning, wrote Python programs that reduced the dataset dimensionality, and numerically encoded protein amino acid sequences. Implemented various machine learning model to predict the protein mutation sites for optimized photostability, response amplitude, and brightness. Presented project progress to laboratory and fellow undergraduates.

Advisors: Jon Clauss, Ph.D. and Andrew Sward, Ph.D.

Description: Investigated introductory topology concepts and constructed example proofs with Dr. Jon Clauss. Presented definitions and examples related to metric spaces and epsilon balls to the Mathematics and Computer Science Department during a campus-wide research presentation day. Collaborated with Dr. Andrew Sward developing mathematical-themed non-fungible tokens to deploy for purchase a trade. Presented proof-of-concept tokens to the Mathematics and Computer Science Department for feedback and guidance.

Funding & Awards

NSF Fellowship: AI and Data Enabled Predictive Multiscale Modeling	2023-2024
Best Poster at MSU Engineering Graduate Research Symposium	2024
Harry Nelson Award for Excellence in Applied Mathematics, Augustana College	2023

Publications

Ross, G., Huang, A. W., **Reiling, J.**, Zhang, M., Park, J., Radtke-Schuller, S., ... Fröhlich, F. Switching state to engage and sustain attention: dynamic synchronization of the frontoparietal network. *Progress in Neurobiology*. 2025. 0301-0082. doi: 10.1016/j.pneurobio.2025.102777.

Selected Conference Presentations

Reiling, J., Frohlich, F., Zhang, M. (2025, November 16). *Topological data analysis reveals stable relative postural states during social interaction between freely moving animals*. (Poster, Society for Neuroscience Annual Meeting, San Diego, CA)

Reiling, J., Frohlich, F., Zhang, M. (2024, October 22) *Topological data analysis characterizes rich behavioral dynamics in naturalistic social interaction in ferrets*. (Poster, NSF Research Traineeship Annual Meeting, Arlington, VA)

Reiling, J., Frohlich, F., Zhang, M. (2024, October 8) *Topological data analysis characterizes rich behavioral dynamics in naturalistic social interaction in ferrets*. (Poster, Society for Neuroscience Annual Meeting, Chicago, IL)

Reiling, J., Zhang, M. (2024, May 2). *Improving accuracy of multi-animal motion tracking by supplementing deep learning with body model-based automatic corrections*. (Best Poster Award, Engineering Graduate Research Symposium, Michigan State University, East Lansing, MI)

Invited Talks

Reiling, J (2023). *Linear algebra in machine learning: eigenvectors and eigenvalues implemented in principal component analysis*. (Celebration of Learning, Augustana College)

Reiling, J (2022). *Machine learning: guided engineering of fluorescent voltage indicators*. (SMART Summer Research Presentation Day, Baylor College of Medicine)

Reiling, J (2022). *Metric spaces and epsilon balls: redefining distance*. (Celebration of Learning, Augustana College)

Reiling, J, Thompson. J, Olana. K, Pham. T (2021). *NFTrig: Trigonometry based non-fungible tokens (NFTs)*. (Illinois Section of the Mathematical Association of America, Milikin University)

Leadership & Membership

- Society for Neuroscience, Student Membership 2024-Present
- President of Graduate Student Organization Executive Board 2025-2026
- CMSE Peer Mentorship Program, Mentor 2024-2026
- Vice President of Graduate Student Organization Executive Board 2024-2025
- Secretary of Graduate Student Organization Executive Board 2024
- Phi Beta Kappa, National Honors Society 2023
- Pi Kappa Lambda, National Music Honors Society 2023
- Pi Mu Epsilon, National Honorary Mathematics Society 2021
- Eagle Scout 2017

Relevant Coursework at Michigan State University

Topological Data Analysis Fall 2025
Description: Critically understand, develop, and implement software using topological data analysis methods including persistence algorithms, filtrations, Betti curves, Rebb graphs, Mapper, multiparameter persistence, and topological signal processing.

Systems & Behavioral Neuroscience I Spring 2025
Description: Studied conceptual foundations and current research in circadian rhythm, neural pathways regulating sleep and arousal, neurobiology of male and female reproductive behavior, motivated behavior and reward circuitry, and neural mechanisms of learning and memory. Developed skills to critically read, write, and orally present scientific research in behavioral neuroscience.

Biomedical Signal Processing Fall 2024
Description: Deterministic and random digital signal processing theory in the context of biomedical applications with hands-on projects about the acquisition and analysis of real physiologic signals. Additional implementation of machine learning techniques for biomedical signals.

Mathematical Foundations of Data Science Spring 2024
Description: Studied the ability to process, extract, and utilize insightful information form large datasets which provides useful knowledge, support decision-making, uncover hidden trends, and enable deeper understanding of observed phenomena.

Parallel Computing

Spring 2024

Description: Explored the benchmark and profile performance of serial and parallel applications, developing and optimizing applications using shared-memory threading parallelism, distributed-memory message passing, and hybrid parallelism to understand the current state of high-performance parallel computing.

Numerical Methods for Differential Equations

Fall 2023

Description: Learned numerical methods of ordinary and partial differential equations, including elliptic, parabolic, and hyperbolic equations, explicit and implicit solutions, in addition to numerical error and stability analysis.

Numerical Linear Algebra

Fall 2023

Description: Studied numerical concepts and methods for efficiently solving linear equations and eigenvalue problems. Major topics include fundamental matrix factorization, solving linear systems, algorithmic analysis, and iterative solvers.

Computational Techniques

- **MATLAB:** Implemented numerical methods of ordinary and partial differential equations using Finite Element Method and Finite Difference Method. Developed scientific programming projects for research-related projects and graduate coursework.
- **Python:** Developed data manipulation methods for automatic time series correction for animal motion tracking research at Michigan State University. Additional training in python projects during undergraduate coursework.
- **MySQL:** Learned and applied MySQL database language to multiple projects and assignments. Implemented MySQL in databases which connected to website user interfaces.
- **Java:** Created optimized college registration program which produces calendar schedule from Augustana College class registration information in Software Development class. Utilized collaborative repository environments, including GitHub and Git bash. Implemented JavaFX objects to develop user-friendly client interfaces.

Mentoring Experience

- **Ab Basit Rafi Syed** (master's student, Data Science, MSU) January 2025-October 2025
Naturalistic social ferret motion tracking with computer vision
- **Vitika Patel** (undergraduate student, Biochemistry, MSU) November 2024-January 2025
Naturalistic social ferret motion tracking with computer vision
- **Shreya Karikrishnan** (undergraduate student, Neuroscience, UNC-CH) May 2024-July 2024
Task-based ferret motion tracking with computer vision

Teaching

Montreal Open Tools Symposium

Tutorial Assistant

Quebec, Canada

September 2025

Taught graduate students how to use Temporal Mapper, a topological data analysis and dynamical systems framework, to extract rhythmic states and their transitions from musical recording.

Augustana College

Undergraduate Tutor

Illinois, U.S.

September 2020-May 2021

Provided instruction and feedback for precalculus, calculus I and II, and multivariable calculus students for fundamental concepts and examples.